A

Internship Report

On

Tree Species Recognition

Submitted by

Kavya Shriniwas Kolamkar

Of T.E (Computer Engineering- Course 2019) having Examination Seat No.

Roll No.TCA07 in partial fulfillment for the a ward of degree Bachelor of Engineering in Computer Engineering for Academic Year 2025-2026

Under the guidance of

Dr. Sandeep Patil

Department of Computer Engineering

Hope Foundation's

International Institute of Information Technology, Hinjewadi,

Pune – 411057

Savitribai Phule Pune University, Pune

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Pune – 411057

Academic Year 2023-2024

Department of Computer Engineering

Certificate

This is to certify that, the Internship Report entitled "***Tree Species Recognition***" submitted by, of Third Year having Roll No. TCA32 , is a bonafide work completed under my supervision and guidance in partial fulfillment for award of degree of Bachelor of Engineering in the branch Computer Engineering of International Institute of Information Technology, Hinjewadi, Pune-57

Prof. Name

Dr. Sandeep Patil

Dr. Ajitkumar Shitole	**Dr. Vaishali V. Patil**
Head of Department	**Principal**

Acknowledgment

It gives me great pleasure in presenting the Internship report on '***Tree Species Recognition***'. I would like to take this opportunity to thank my internship mentor **Dr. Sandeep Patil Sir** for giving me all the help and guidance I needed. I am grateful to him/her for his/ her kind support. His/ Her valuable suggestions were very helpful.

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At last, I must express our sincere heartfelt gratitude to all the faculty members of the Computer Engineering Department who helped me directly or indirectly during this project work.

Kavya Shriniwas Kolamkar

Student name

CLASS:- **TE(**Computer Engineering) ROLL NO. :- **TCA-07**

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Company Details-

- Name- Forest By Heartfulness
- Address- HEARTFULNESS INSTITUTE
  D NO.13-110, Kanha Village, Nandigama
  Mandal, Ranga Reddy district, Telangana
  - 509325, India.
- Supervisor Name- Mr. Ved Mone
- Contact No- +91 8237604320
- Email Id- vedmone9@gmail.com
- Internship domain- *Machine Learning* | *Computer Vision*
- Duration Of Internship - 6 month
- Nature of Work- Remote (online)
- Are they providing placement – No

1.Abstract

This project presents a Human-in-the-Loop training system with a Man-in-the-Middle interface designed to improve classification performance through iterative review and retraining. The system integrates a frontend review interface, backend training pipeline, and automated deployment mechanism to create a continuous learning workflow. Initially, predictions generated by the model are presented to a human reviewer through the interface, where incorrect or low-confidence predictions are manually validated and corrected. These reviewed samples are then used to retrain a Support Vector Machine (SVM) model, improving classification accuracy over time.

After retraining, the updated model is automatically deployed and used to re-score the dataset. The system prioritizes samples with the lowest confidence scores, enabling targeted human intervention where the model is most uncertain.

This iterative cycle of :

Review → Train → Deploy → Predict → Repeat

This ensures progressive model improvement while minimizing manual effort. The architecture combines machine learning automation with human intelligence, resulting in a scalable and efficient training workflow. The implemented solution includes a user-friendly frontend, backend processing pipeline, and automated retraining loop, demonstrating an effective approach for improving model performance in real-world scenarios.

2. Objectives

- To design and develop a Human-in-the-Loop training system for improving model prediction accuracy.
- To create a Man-in-the-Middle review interface that allows manual validation and correction of model predictions.
- To implement a frontend application for displaying predictions and collecting human feedback efficiently.
- To develop a backend pipeline that processes reviewed data and prepares it for retraining.
- To train a Support Vector Machine (SVM) model using corrected samples obtained from human review.
- To automate model deployment after retraining for continuous system improvement.
- To prioritize low-confidence predictions for review to reduce manual effort and improve learning efficiency.
- To build an iterative workflow consisting of Review → Train → Deploy → Predict → Repeat.
- To improve classification performance over time using incremental retraining.
- To demonstrate an end-to-end system integrating user interface, backend processing, and machine learning workflow.

3.

Introduction

Machine learning models often require large amounts of accurately labeled data to achieve high performance. However, in real-world scenarios, predictions generated by automated systems may contain errors, especially when the model encounters new or uncertain data. Relying solely on automated training can lead to reduced accuracy and poor generalization. To address this challenge, Human-in-the-Loop (HITL) systems are introduced, where human expertise is incorporated into the training process to improve model performance iteratively.

This project presents a Human-in-the-Loop training workflow implemented through a Man-in-the-Middle interface that connects model predictions with human validation. The system allows users to review model outputs, correct incorrect predictions, and feed the validated data back into the training pipeline. By integrating manual review with automated learning, the system ensures that the model continuously improves over time.

The proposed workflow consists of five major stages: Review, Train, Deploy, Predict, and Repeat. Initially, predictions generated by the model are displayed in the frontend interface for human review. The reviewed and corrected samples are then passed to the backend training pipeline, where a Support Vector Machine (SVM) model is retrained. Once retraining is completed, the updated model is automatically deployed. The deployed model then re-scores the dataset, and samples with the lowest confidence are prioritized for the next review cycle. This iterative loop continues until the desired performance level is achieved.

The system also includes a user-friendly frontend interface for manual validation, a backend module for data processing and retraining, and an automated deployment mechanism. By combining human intelligence with machine learning automation, the proposed solution reduces labeling effort, improves prediction accuracy, and enables continuous model improvement. This approach is particularly useful in scenarios where labeled data is

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limited and model performance needs to improve progressively through guided human feedback.

Design and Mehodology

The proposed system follows a Human-in-the-Loop architecture with a Manin-the-Middle interface that connects model predictions, human validation, and automated retraining. The overall design consists of three major components: Frontend Review Interface, Backend Training Pipeline, and Automated Deployment Module. These components work together in an iterative workflow to continuously improve model performance.

System Architecture

The workflow of the system follows the sequence:

Review (Human) → Train (SVM) → Deploy (Auto) → Predict (Re-score) → Repeat

In the first stage, predictions generated by the model are displayed in the frontend interface. The reviewer manually verifies each prediction and corrects any misclassified samples. The corrected labels are stored and passed to the backend for further processing.

1. Review Module (Human-in-the-Loop Interface)

The review module acts as a Man-in-the-Middle interface between prediction and training. It displays model predictions along with confidence scores. The user can accept or correct predictions. The system prioritizes samples with the lowest confidence values so that human effort is focused on uncertain cases. Once reviewed, the validated data is saved and sent to the training module.

2. Training Module (SVM Retraining)

After collecting reviewed samples, the backend training pipeline prepares the dataset for retraining. Features are extracted from the reviewed data and combined with previously labeled samples. A Support Vector Machine

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(SVM) model is then retrained using the updated dataset. This incremental retraining improves classification performance based on human feedback.

3. Deployment Module (Automatic Update)

Once the training process is completed, the updated model is automatically deployed. The deployment module replaces the previous model with the newly trained one without manual intervention. This ensures that the system always uses the most recent and improved model.

4. Prediction and Re-scoring Module
After deployment, the model re-runs predictions on the dataset. Each prediction is assigned a confidence score. The system then ranks samples based on confidence values and selects the lowest confidence predictions for the next review cycle.

5. Iterative Learning Loop
The entire system follows an iterative learning approach. The cycle of Review → Train → Deploy → Predict continues until the model achieves stable performance. This methodology reduces manual labeling effort while continuously improving model accuracy.

Methodology Summary

Step 1: Generate predictions using the trained model
Step 2: Display predictions in review interface
Step 3: Human validates and corrects predictions
Step 4: Store reviewed samples
Step 5: Retrain SVM model using updated dataset
Step 6: Automatically deploy updated model
Step 7: Re-score dataset using new model
Step 8: Select lowest confidence samples
Step 9: Repeat the process

This design ensures continuous learning, efficient use of human feedback, and progressive model improvement.

5. Implementation

Implementation

The proposed Human-in-the-Loop training system was implemented using a frontend review interface, backend processing pipeline, and an automated model retraining workflow. The implementation integrates manual validation with machine learning to enable continuous improvement of model predictions.

Frontend Implementation (Review Interface)

The frontend interface was developed to display model predictions along with their confidence scores. The interface acts as a Man-in-the-Middle system where users can review, accept, or correct predicted labels. Each sample is presented with necessary information, and the reviewer can modify incorrect predictions. The interface also prioritizes samples with the lowest confidence values to ensure that human effort is focused on uncertain predictions. Once reviewed, the corrected labels are sent to the backend through API calls.

Backend Implementation

The backend handles data processing, storage of reviewed samples, retraining, and deployment. When reviewed data is received from the frontend, the backend updates the dataset and prepares it for retraining. Feature extraction is performed on the validated samples, and the updated dataset is combined with previously labeled data. The backend then triggers the training pipeline.

Model Training (SVM)

A Support Vector Machine (SVM) classifier was used for model training. The retraining process uses corrected samples collected from the review interface. The model is trained incrementally using the updated dataset to improve classification accuracy. After training, the model is saved and made ready for deployment.

Automatic Deployment
Once retraining is completed, the system automatically deploys the updated model. The previously deployed model is replaced with the newly trained model without manual intervention. This ensures that the system always uses the most recent version of the model for predictions.

Prediction and Re-scoring
After deployment, the updated model generates predictions on the dataset. Each prediction is assigned a confidence score. The backend ranks predictions based on confidence values and selects the lowest confidence samples. These samples are then sent back to the frontend review interface for manual validation.

Iterative Workflow Execution
The system continuously runs in an iterative loop:

1. Model generates predictions
2. Predictions shown in frontend review interface
3. Human validates and corrects labels
4. Backend updates dataset
5. SVM model retrained
6. Updated model automatically deployed
7. Dataset re-scored using new model
8. Lowest confidence samples selected
9. Process repeated

This implementation enables continuous learning, reduces manual labeling effort, and improves model accuracy over time by combining human feedback with automated machine learning retraining.

6. Input And Output

Input

The system takes model-generated predictions along with confidence scores as input. These predictions are displayed in the frontend review interface for human validation. Each input sample contains the predicted label, confidence value, and associated data instance. The reviewer examines the predictions and corrects any misclassified samples. The validated inputs are then forwarded to the backend for retraining.

The input to the system includes:
• Model predictions generated from the current trained model
• Confidence scores for each prediction
• Data samples displayed in review interface
• Human-corrected labels from the frontend
• Previously labeled training dataset

The system prioritizes samples with the lowest confidence scores to improve learning efficiency.

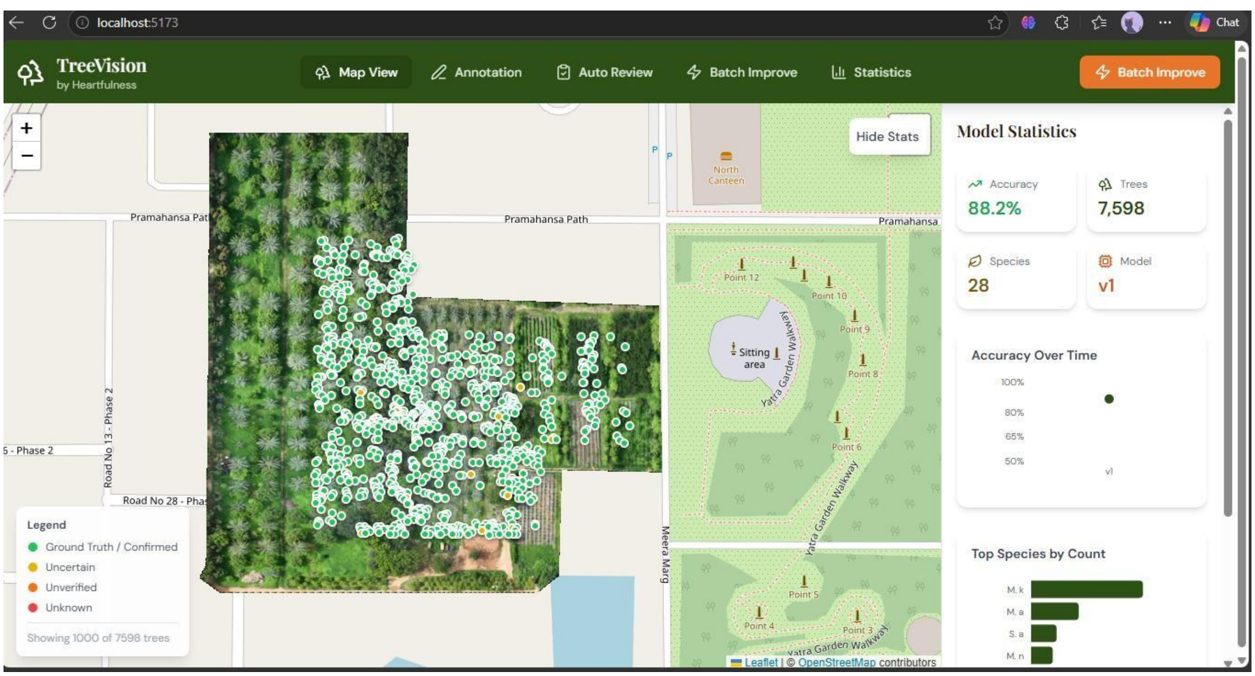

Fig 1: Tree Vision Map View

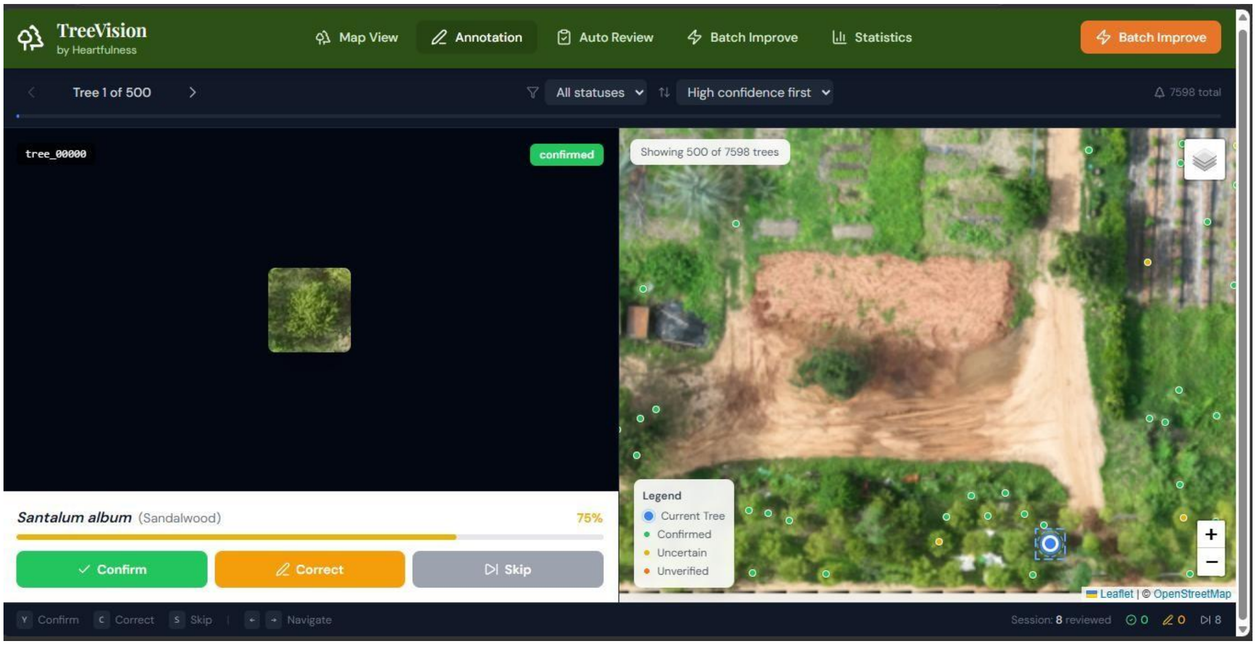

Fig 2: Annotation Tab

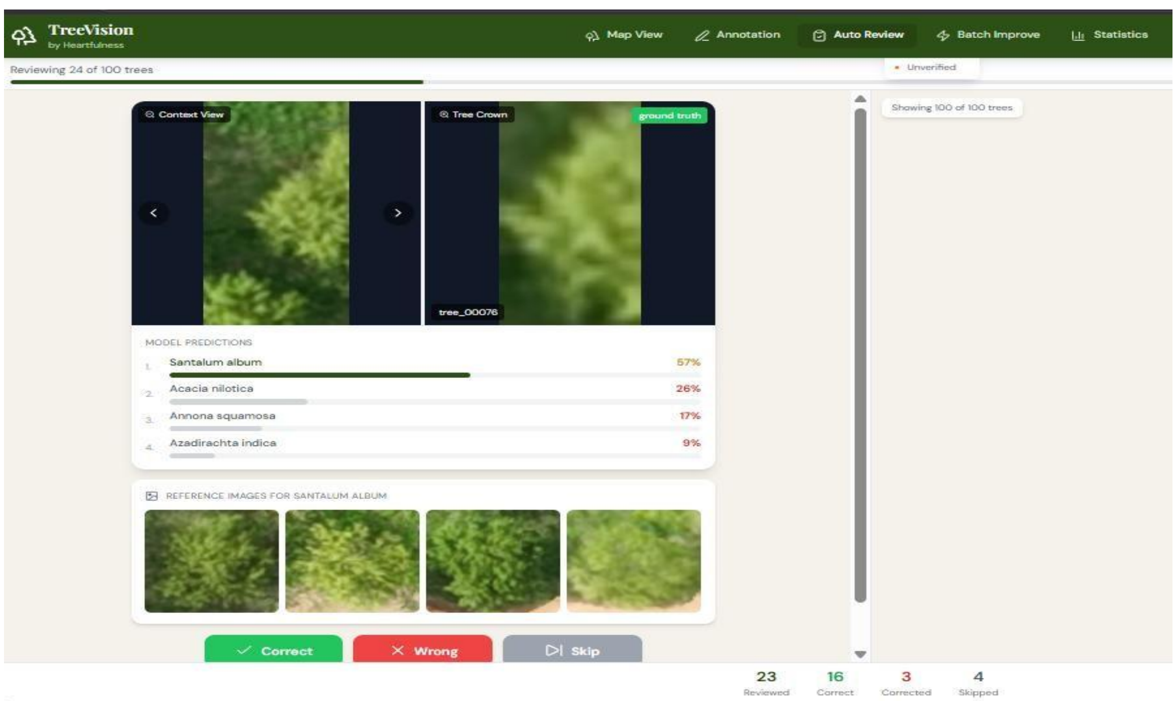

Fig 3: Auto Review

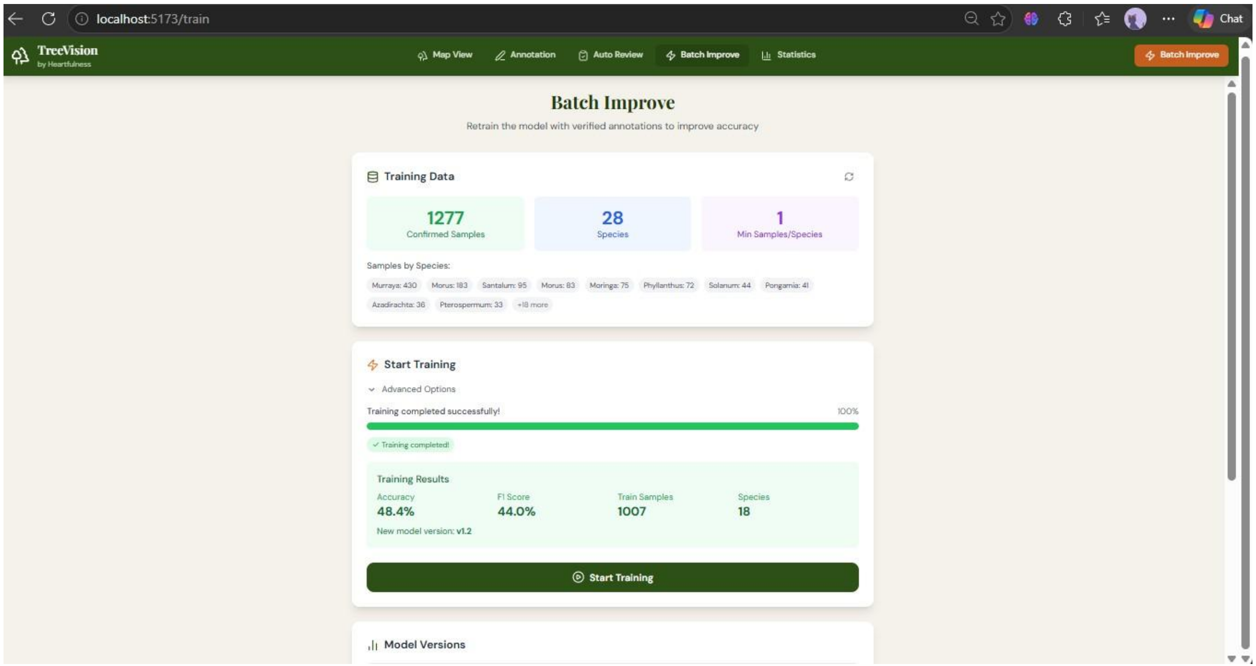

Fig 4: Batch Improvement

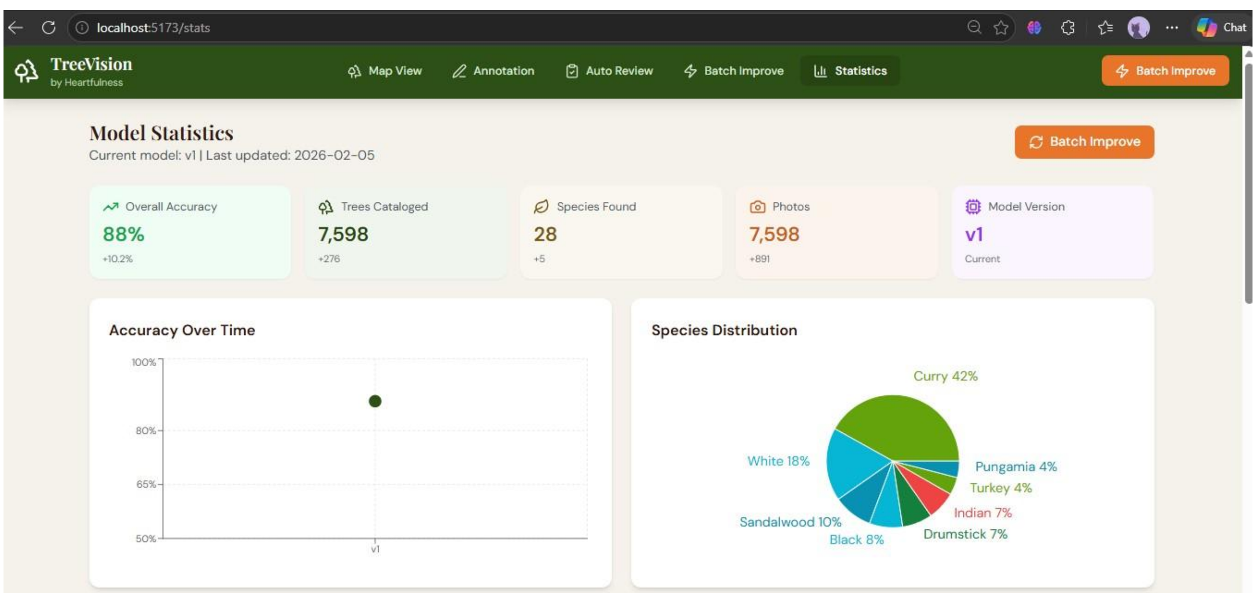

Fig 5: Statistics

Output (Results)

The system produces corrected labels, retrained model outputs, and updated predictions after each iteration. Once the human review is completed, the corrected samples are used to retrain the Support Vector Machine (SVM) model. The updated model is then deployed automatically and used to rescore the dataset.

The output generated by the system includes:
• Corrected labels from human review
• Retrained SVM model
• Updated predictions using new model
• Confidence scores for each prediction
• Ranked list of lowest confidence samples
• Improved classification accuracy over iterations

The results demonstrate continuous improvement in model performance as the review and retraining cycle progresses.

7. Include Demo Video

https://youtu.be/U31fl_jEmQw?si=lCAOCte7T9enC7Xc

8. Future Enhancement

The proposed Human-in-the-Loop training system can be further enhanced to improve scalability, automation, and model performance. Several improvements can be incorporated in future versions of the system to make it more efficient and robust.

- Further improvement of model accuracy by incorporating additional reviewed samples and iterative retraining cycles.
- Integration of advanced machine learning and deep learning models to enhance classification performance on complex datasets.
- Implementation of Active Learning strategies to automatically select the most informative samples for review.
- Batch review functionality to allow users to validate multiple samples simultaneously and reduce manual effort.
- Real-time or incremental training to update the model without complete retraining.
- Multi-user review system where multiple reviewers can validate predictions collaboratively.
- Extension of the system to track growth and changes of large numbers of trees over time using temporal data. This would enable monitoring of tree health, density, and development across multiple time periods.
- Integration with geospatial data to analyze tree distribution and enable large-scale environmental monitoring.
- Deployment as a scalable web-based application capable of handling large datasets and continuous updates.

These enhancements would further improve model accuracy, enable largescale tree monitoring, and make the system more efficient for long-term analysis and decision-making.

9.Conclusion

The project successfully implements a Human-in-the-Loop training system with a Man-in-the-Middle interface for improving model predictions. The workflow integrates human review, SVM retraining, automatic deployment, and re-scoring in an iterative loop. By prioritizing low-confidence samples for validation, the system reduces manual effort and continuously improves model accuracy. The developed solution provides an efficient and scalable approach for model refinement and can be further extended for large-scale tree monitoring and accuracy enhancement in future.

10.Web Research Link

1. Human-in-the-Loop Machine Learning https://cloud.google.com/architecture/human-in-the-loop-ml
2. Active Learning and Human-in-the-Loop Overview https://www.ibm.com/topics/human-in-the-loop
3. Support Vector Machine (SVM) – Scikit-learn Documentation https://scikit-learn.org/stable/modules/svm.html
4. Active Learning Concepts https://www.geeksforgeeks.org/active-learning-machine-learning/
5. Confidence-Based Sampling in Machine Learning https://towardsdatascience.com/active-learning-sampling-strategies1b7d7c6b5a2
6. Human-in-the-Loop AI Systems https://www.microsoft.com/en-us/research/project/human-aiinteraction/
7. Machine Learning Model Deployment Concepts https://aws.amazon.com/what-is/model-deployment/
8. Iterative Model Training and Feedback Loops https://neptune.ai/blog/ml-model-retraining
9. SVM Classification Explanation https://www.geeksforgeeks.org/support-vector-machine-algorithm/
10. Active Learning for Image Classification https://labelbox.com/blog/active-learning-with-machine-learning/

11.References

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11. TensorFlow, "Human-in-the-Loop Pipelines," Available: https://www.tensorflow.org/tfx/guide

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13. Kaggle Learn, “Machine Learning Model Improvement,” Available: https://www.kaggle.com/learn/intro-to-machine-learning

14. Stanford University, “Support Vector Machines Explained,” Available: https://cs229.stanford.edu/notes2020fall/cs229-notes3.pdf

15. ResearchGate, “Human-in-the-Loop Machine Learning Systems,” Available: https://www.researchgate.net/publication/324930234_Human-in-theLoop_Machine_Learning

12. Include your 2 photos of industry working environment

Date: 1 January 2026

Subject: Internship Offer

Dear **Kavya Shriniwas Kolamkar**,

This is to inform you that **Forests by Heartfulness** is offering you an **internship** in the role of **AI/ML Project Intern**.

The internship will be for a duration of **January to May** (subject to change).
Mode of internship: **Remote**.
Stipend (if applicable): **Not applicable**.

You are expected to follow the rules and guidelines of the organization during the internship period. A certificate will be issued upon successful completion.

Please confirm your acceptance of this offer.

Regards,

Mr. Chakradhar Gandhe
Country Head, Program Management
Forests by Heartfulness
chakradhar.gandhe@heartfulness.org

14. Internship Completion Letter